Project Report

Fashion Rental Store

Course:

DATABASE MANAGEMENT SYSTEM

YEAR: 2025

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I. Storyline

In today's fast-paced and style-conscious world, fashion is no longer just about owning—it's about experiencing. Whether it's a designer gown for a gala or a sharp tuxedo for a special evening, people are increasingly embracing fashion rentals as a smarter, more sustainable alternative to traditional shopping. To meet this evolving trend, we created **RENTCOUTURE**, an online platform that allows users to discover, rent, and enjoy high-end fashion without the commitment of ownership.

The system supports two primary user roles: **renters** and **providers**. Renters can browse fashion items by category, filter by event type, size, or brand, and reserve their favorite pieces for a chosen date range. Providers, on the other hand, can upload and manage their wardrobe inventory—adding images, sizes, rental pricing, availability calendars, and care instructions.

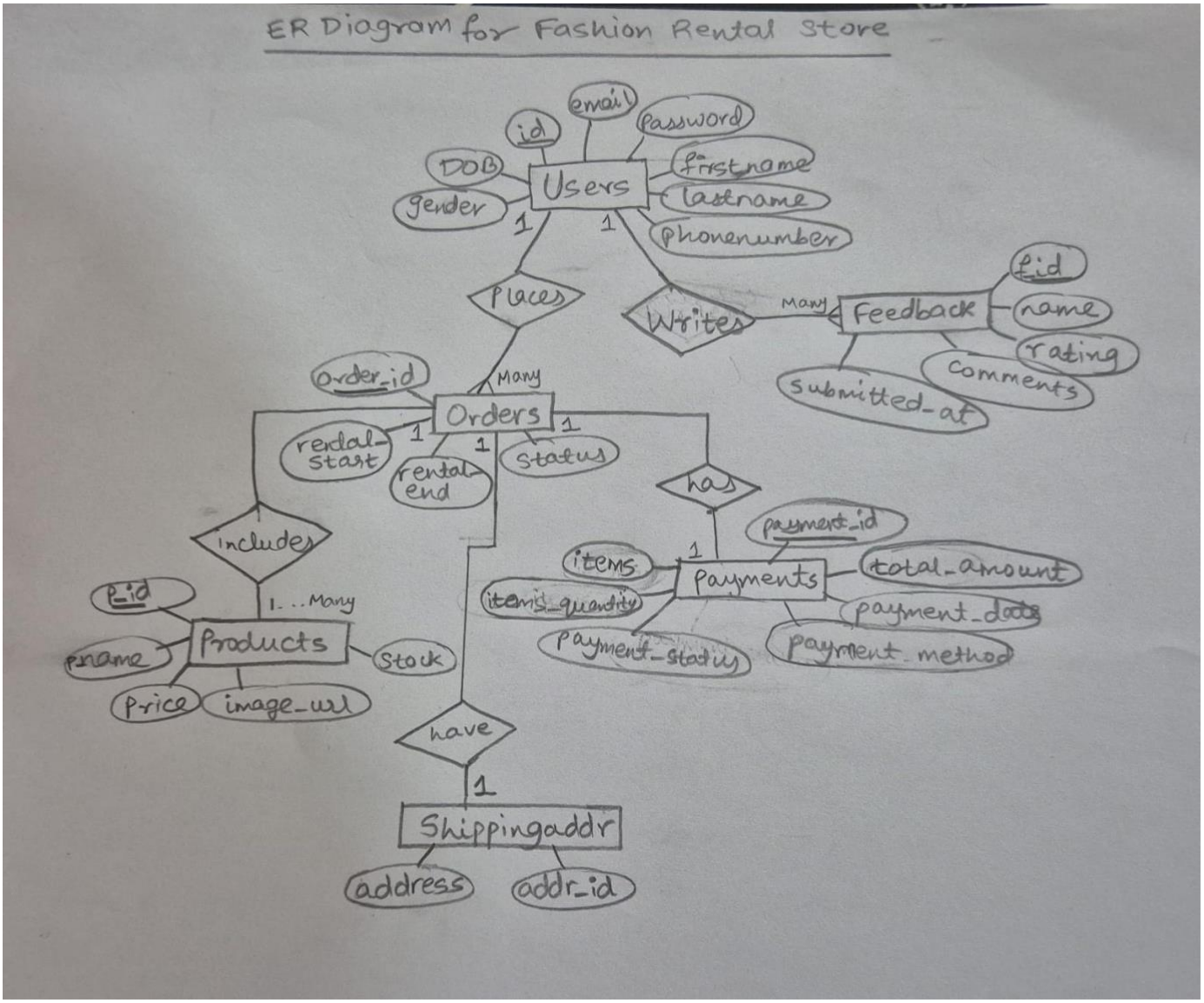
The backend is built using **PHP** and runs on a **XAMPP** server stack, with a **MySQL database** handling the core data. This includes user profiles, fashion items, rental bookings, availability slots, and wallet transactions. **RENTCOUTURE** is designed to be user-friendly, scalable, and a real-world solution for the modern, eco-conscious fashion enthusiast.

II. Components of Database Design

The database was designed to support various functionalities of the fashion rental platform. It includes the following main components:

- **Users Table**: Stores user information such as user ID, email, password, first name, last name, phone number, gender, and date of birth.
- **Products Table**: Contains fashion item details including product ID, product name, rental price, image URL, and available stock.
- **Feedback Table**: Records user feedback with fields like feedback ID, name, rating, comments, and the timestamp of submission.
- **Payments Table**: Captures payment-related data such as payment ID, total amount, payment date, payment method, payment status, quantity of items rented, and item details.
- **Shipping Address Table**: Stores shipping address details with fields like address and address ID, used for delivery of rented items.
- **Orders Table**: Manages rental orders with fields such as order ID, rental start date, rental end date, and current order status.

III. Entity Relationship Diagram



IV. Relational Model

Convert the ER diagram to the relational model using the concepts learned in the class.
List the various tables obtained.
Entity Set Table with attributes, types, and descriptions

1. **Explanation** 2. **Final Relational Schema**

Conversion of ER Diagram to Relational Model

Name | Entity Set / Relationship | Type

Name	Type
Users	Entity Set
Products	Entity Set
Feedback	Entity Set
Payments	Entity Set
Shippingaddr	Entity Set
Orders	Entity Set
Order_Product	Relationship Set
Gives_Feedback	Relationship Set
Makes_Payment	Relationship Set
Places_Order	Relationship Set
Ships_To	Relationship Set

Entity Set: Users

Attributes	Attribute Type	Description
user_id	Primary key	Unique identifier for user
email	Simple	Email of the user
password	Simple	Password
first_name	Simple	First name of the user
last_name	Simple	Last name of the user
phone_number	Simple	Contact number
gender	Simple	Gender of the user
dob	Simple	Date of birth

Explanation

• Strong entity

• Converted to relation:

`Users(user_id, email, password, first_name, last_name, phone_number, gender, dob)`

• `user_id` is Primary Key

• All other attributes are simple

Entity Set: Products

Attributes	Attribute Type	Description
product_id	Primary key	Unique ID of product
product_name	Simple	Name of the fashion item
price	Simple	Rental price
image_url	Simple	URL of the product image
stock	Simple	Quantity available for rent

Explanation

• Strong entity

• Converted to relation:

`Products(product_id, product_name, price, image_url, stock)` • `product_id` is Primary Key

Entity Set: Feedback

Attributes	Attribute Type	Description
feedback_id	Primary key	Unique ID of feedback
user_id	Foreign key	References user who gave it
name	Simple	Name of the user (optional)
rating	Simple	Star rating (e.g., 1–5)
comments	Simple	Textual feedback
submitted_at	Simple	Date and time of submission

Explanation

• Strong entity

• Converted to relation:

`Feedback(feedback_id, user_id, name, rating, comments, submitted_at)`

• `feedback_id` is Primary Key

• `user_id` is a Foreign Key referencing `Users`

Entity Set: Payments

Attributes	Attribute Type	Description
payment_id	Primary key	Unique ID for the payment
order_id	Foreign key	References order being paid
total_amount	Simple	Total amount paid

payment_date	Simple	Date of payment
payment_method	Simple	Method used (Card, Wallet etc)

Attributes Attribute Type Description payment_status Simple Success, Failed, Pending items_quantity Simple Number of items in the order

Explanation

• Strong entity
• Converted to relation:
`Payments(payment_id, order_id, total_amount, payment_date, payment_method, payment_status, items_quantity)`
• `payment_id` is Primary Key
• `order_id` is a Foreign Key from `Orders`

Entity Set: Shippingaddr

Attributes Attribute Type Description address_id Primary key Unique address ID user_id Foreign key References the user address Simple Shipping address string

Explanation

• Strong entity
• Converted to relation:
`Shippingaddr(address_id, user_id, address)`
• `address_id` is Primary Key
• `user_id` is a Foreign Key from `Users`

Entity Set: Orders

Attributes	Attribute Type	Description
order_id	Primary key	Unique ID of the order
user_id	Foreign key	User who placed the order
address_id	Foreign key	Address where order is shipped
rental_start	Simple	Start date of rental period
rental_end	Simple	End date of rental period
status	Simple	Status of the order (e.g. Active)

Explanation
• Strong entity
• Converted to relation:
`Orders(order_id, user_id, address_id, rental_start, rental_end, status)`
• `order_id` is Primary Key
• `user_id`, `address_id` are Foreign Keys

Relationship Set: Order_Product (Many-to-Many)

Attributes Attribute Type **Description** order_id

Foreign key References Orders product_id Foreign key

References Products

quantity Simple Quantity of this product in the order

Explanation
• Many-to-Many relationship
• Converted to relation:
`Order_Product(order_id, product_id, quantity)`
• Composite Primary Key: (`order_id`, `product_id`)

Final Relational Schema

1. **Users**
   `(user_id PK, email, password, first_name, last_name, phone_number, gender, dob)`
2. **Products**
   `(product_id PK, product_name, price, image_url, stock)`
3. **Feedback**
   `(feedback_id PK, user_id FK, name, rating, comments, submitted_at)`
4. **Payments** `(payment_id PK, order_id FK, total_amount, payment_date, payment_method, payment_status, items_quantity)`
5. **Shippingaddr**
   `(address_id PK, user_id FK, address)`
6. **Orders**
   `(order_id PK, user_id FK, address_id FK, rental_start, rental_end, status)`
7. **Order_Product**
   `(order_id FK, product_id FK, quantity)`
   Composite Primary Key: `(order_id, product_id)`

V. Normalization

Perform normalization (1NF, 2NF, 3NF, BCNF) as applicable for the entire database.

1NF (Eliminate Repeating Groups) ●
It has no Composite Values.

2NF (Eliminate Partial Dependency) ●
No partial Dependency.

3NF (Eliminate Transitive Dependency)
● No Transitive property.(A—> B —> C)

BCNF (Boyce-Codd Normal Form)
● All functional dependencies have determinants as a candidate key.

Therefore All the Tables are in Normalized form.

VI. SQL Queries

Using a DBMS software (SQLite3 or MySQL or any other of your choice): ●
Create the tables

1. Table

```
CREATE TABLE Users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
email VARCHAR(100) NOT NULL UNIQUE,
password VARCHAR(100) NOT NULL,
first_name VARCHAR(50),     last_name
VARCHAR(50),     phone_number VARCHAR(15),
    gender ENUM('Male', 'Female', 'Other'),
dob DATE );
```

---

2. Products Table

```
CREATE TABLE Products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
product_name VARCHAR(100) NOT NULL,     price
```

```
DECIMAL(10, 2) NOT NULL,     image_url TEXT,
stock INT DEFAULT 0 );
```

---

3. Feedback Table

```
CREATE TABLE Feedback (
    id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,     name
VARCHAR(100) NOT NULL,
    rating INT NOT NULL CHECK (rating BETWEEN 1 AND 5),
comments TEXT NOT NULL,
    submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (id) REFERENCES Users(id) );
```

---

4. Payments Table

```
CREATE TABLE Payments (
    payment_id INT AUTO_INCREMENT PRIMARY KEY,
order_id INT,
    total_amount DECIMAL(10, 2),
    payment_date DATETIME DEFAULT CURRENT_TIMESTAMP,
payment_method ENUM('Card', 'Wallet', 'Cash', 'UPI'),
payment_status ENUM('Success', 'Pending', 'Failed'),
items_quantity INT,
    FOREIGN KEY (order_id) REFERENCES Orders(order_id) );
```

---

5. Shippingaddr Table

```
CREATE TABLE Shippingaddr (
    address_id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT,
    address TEXT NOT NULL,
    FOREIGN KEY (user_id) REFERENCES Users(user_id) );
```

---

6. Orders Table

```
CREATE TABLE Orders (
    order_id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT,     address_id INT,
rental_start DATE,     rental_end DATE,
    status ENUM('Active', 'Completed', 'Cancelled', 'Pending'),
    FOREIGN KEY (user_id) REFERENCES Users(user_id),
    FOREIGN KEY (address_id) REFERENCES Shippingaddr(address_id) );
```

This section should contain the question, SQL code, and the output snapshot for each query.

SQL Queries :

1. **Fetch all users**

```
SELECT * FROM Users;
```

2. **Get all female users born before 2005**

```
SELECT * FROM Users WHERE gender = 'Female' AND dob < '2005-01-01';
```

3. **Get names and emails of users whose first name starts with 'S'**

```
SELECT first_name, email FROM Users WHERE first_name LIKE 'S%';
```

4. **Count users grouped by gender**

```
SELECT gender, COUNT(*) AS total_users FROM Users GROUP BY gender;
```

5. **List users who haven't placed any orders**

```
SELECT * FROM Users WHERE user_id NOT IN (SELECT DISTINCT user_id FROM Orders);
```

6. **Fetch all products with stock less than 5**

```
SELECT * FROM Products WHERE stock < 5;
```

7. **Products with price between 500 and 1000**

```
SELECT product_name, price FROM Products WHERE price BETWEEN 500 AND 1000;
```

8. **Top 5 most expensive products**

```
SELECT * FROM Products ORDER BY price DESC LIMIT 5;
```

9. **List all out-of-stock products**

```
SELECT * FROM Products WHERE stock = 0;
```

10. **Search for products that contain "dress" in the name**

```
SELECT * FROM Products WHERE product_name LIKE '%dress%';
```

11. **Get all completed orders for user_id = 1**

```
SELECT * FROM Orders WHERE user_id = 1 AND status = 'Completed';
```

12. List orders that started in the last 7 days

```
SELECT * FROM Orders WHERE rental_start >= CURDATE() - INTERVAL 7 DAY;
```

13. Count orders by status

```
SELECT status, COUNT(*) AS total_orders FROM Orders GROUP BY status;
```

14. Get the total amount spent by each user from Payments

```
SELECT o.user_id, SUM(p.total_amount) AS total_spent FROM Payments p JOIN Orders o ON
p.order_id = o.order_id GROUP BY o.user_id;
```

15. Average feedback rating per user

```
SELECT user_id, AVG(rating) AS avg_rating FROM Feedback GROUP BY user_id;
```

16. Get the latest 5 payments made (by date)

```
SELECT * FROM Payments ORDER BY payment_date DESC LIMIT 5;
```

17. Show the most frequently used payment method

```
SELECT payment_method, COUNT(*) AS method_usage FROM Payments GROUP BY payment_method
ORDER BY method_usage DESC LIMIT 1;
```

18. Show feedback with ratings less than 3

```
SELECT * FROM Feedback WHERE rating < 3;
```

19. Get all addresses associated with user_id = 2

```
SELECT address FROM Shippingaddr WHERE user_id = 2;
```

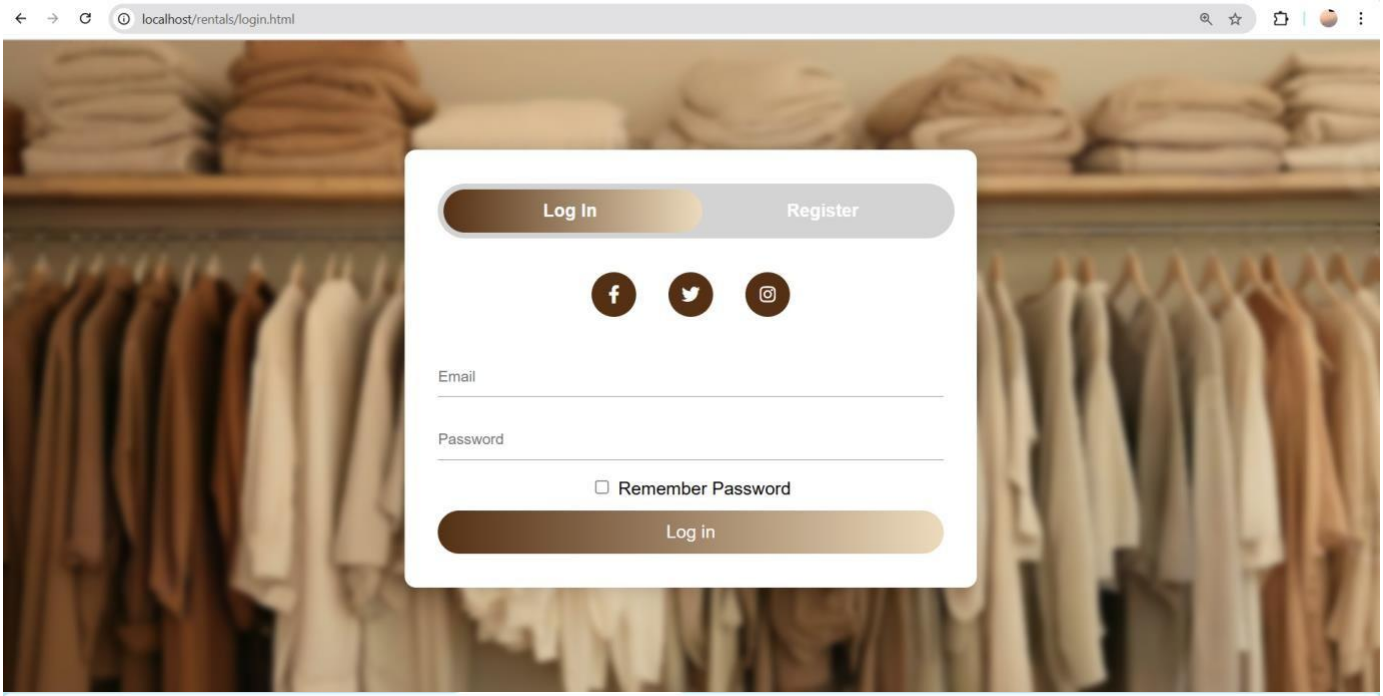
20. Count of orders placed from each address

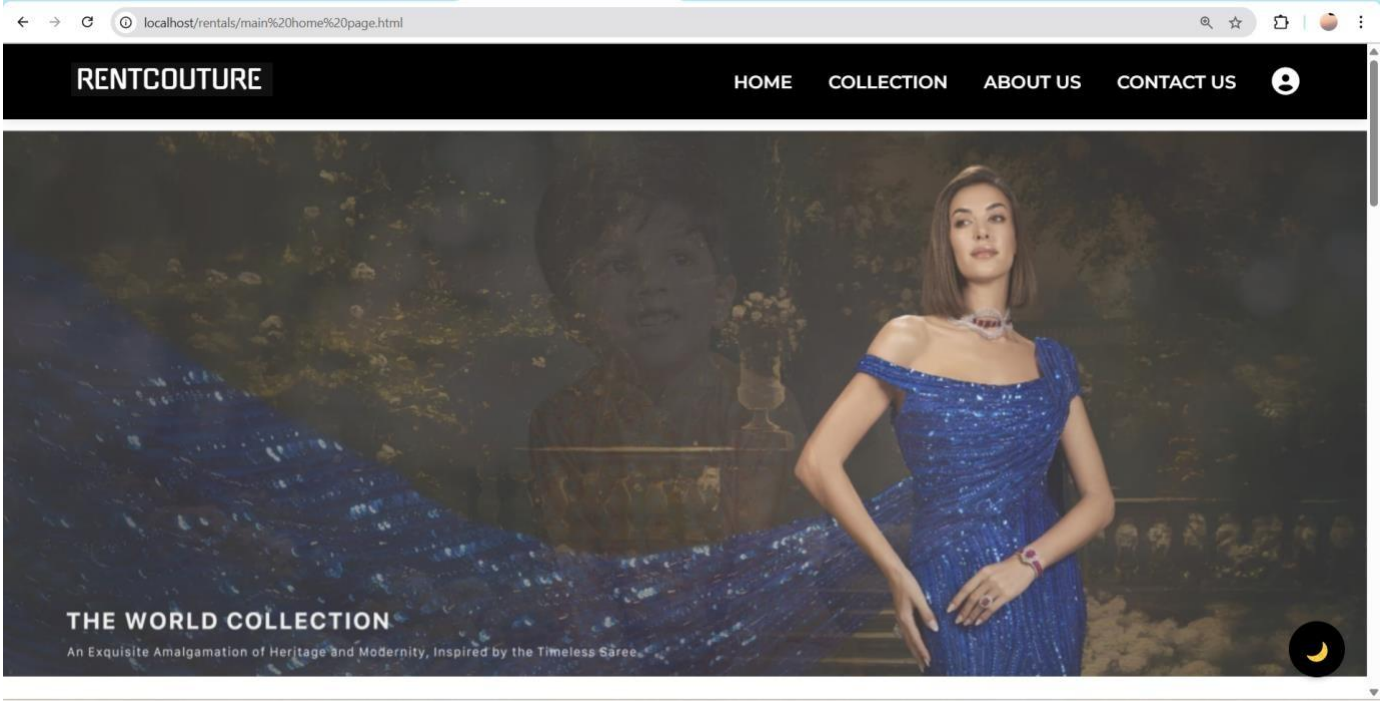
```
SELECT address_id, COUNT(*) AS order_count FROM Orders GROUP BY address_id;
```

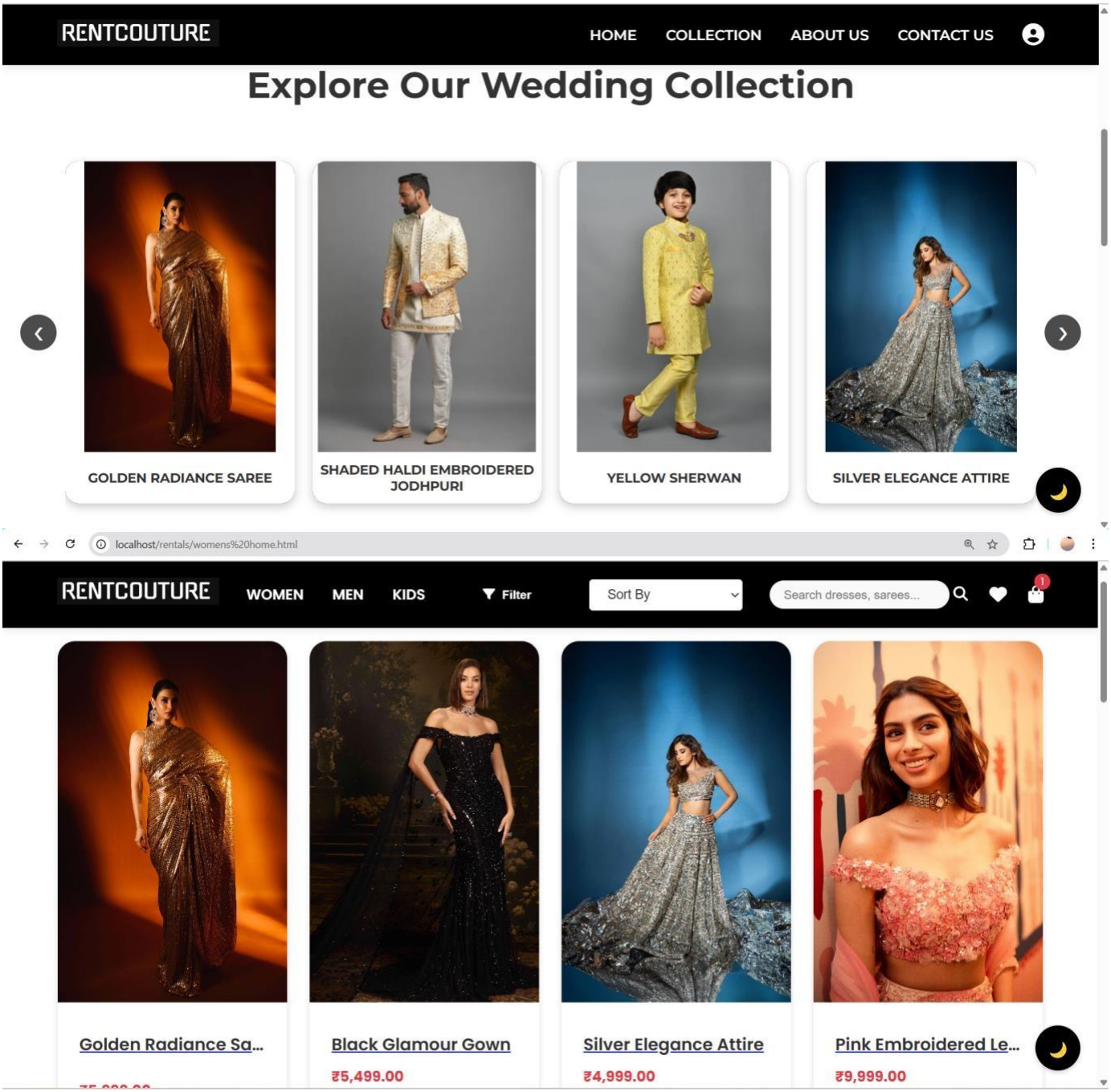
VI. Project demonstration

Tech Stack

Frontend: HTML, CSS, JavaScript
Backend: PHP, XAMPP (for local development environment) Database: MySQL







VII. Self -Learning beyond classroom

This project provided us with a valuable opportunity to apply theoretical concepts in a practical environment. By working on a real-world web application like **RentCouture**, a fashion rental platform, we gained hands-on experience in developing a fully functional online system. Below are some of the key learnings from this

Your Cart

Golden Radiance Saree

₹5,999.00

Silver Elegance Attire

₹4,999.00

Total: ₹16,497.00

Proceed to Checkout →

Checkout

Enter your shipping information

Contact Information

Email *

sachi@gmail.com

Phone *

Shipping Address

- (●) Same as billing address
- ( ) Use a different shipping address

First Name *

Last Name *

Address *

Apartment, suite, etc. (optional)

Country *

Select Country

City *

State/Province *

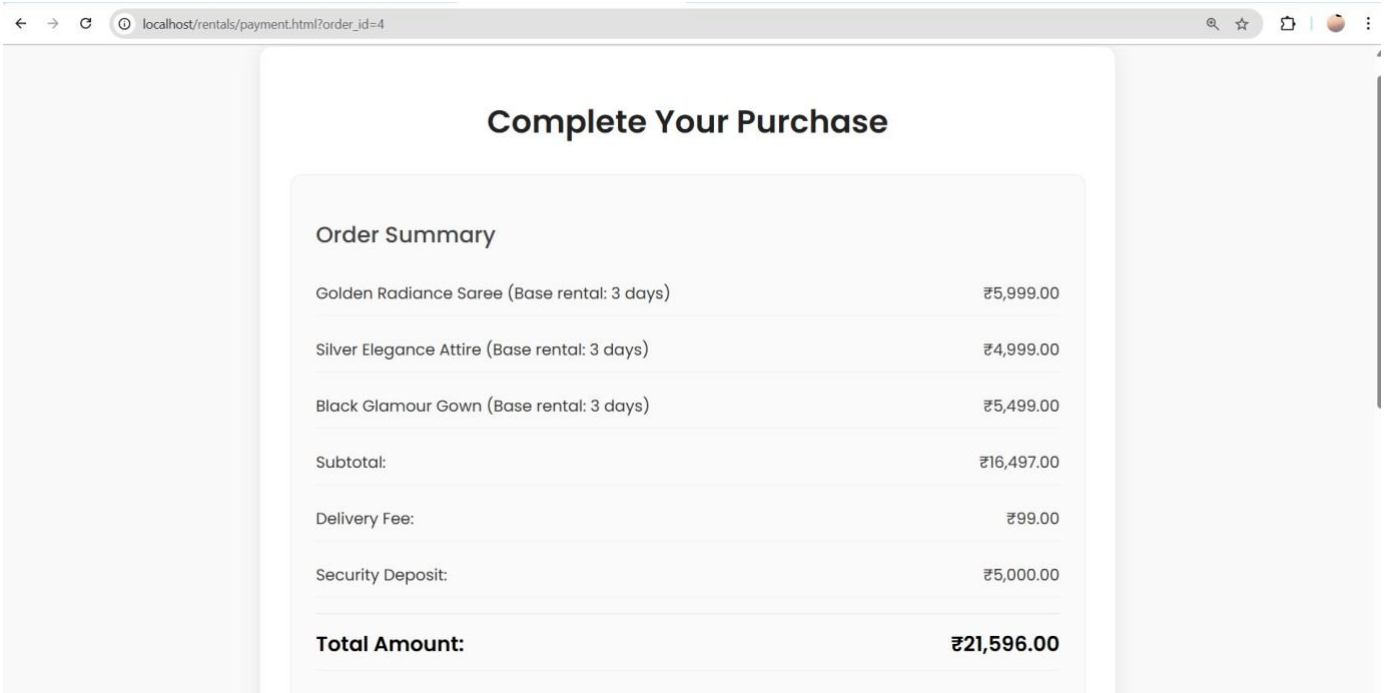
Select State

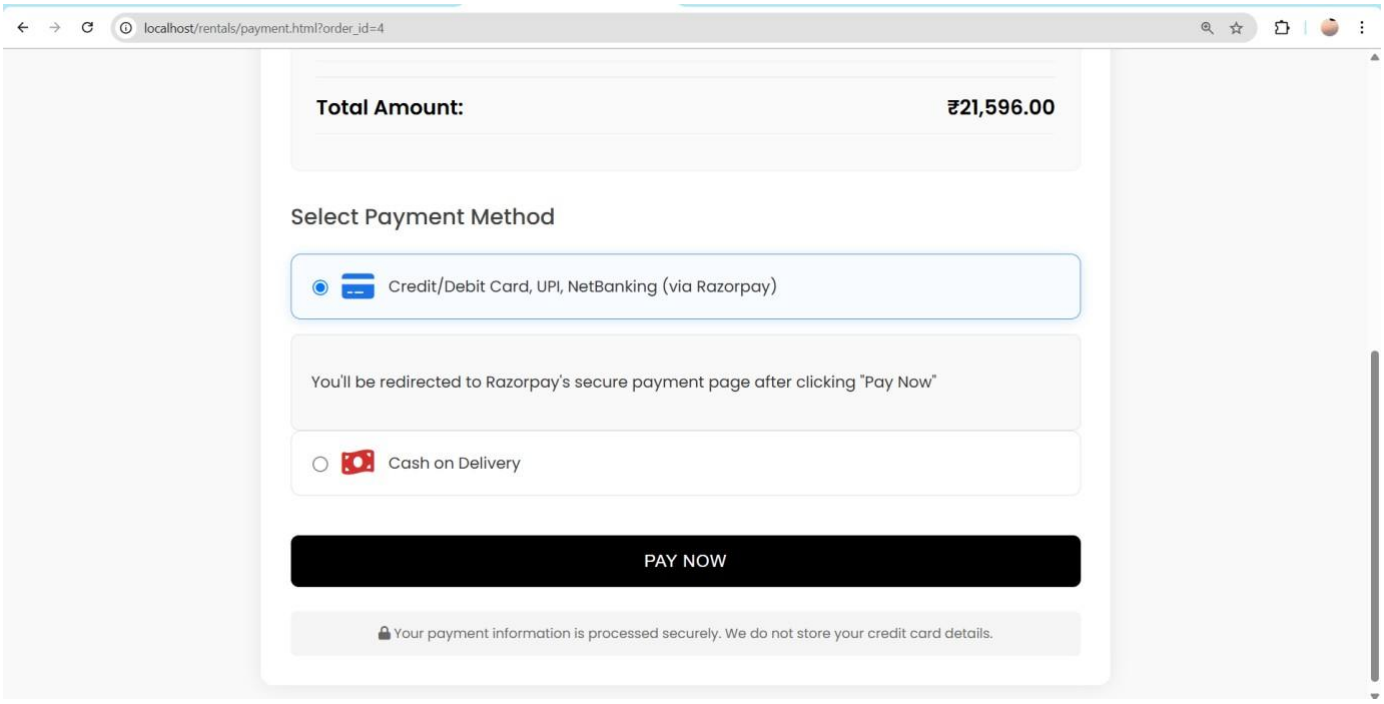
ZIP/Postal Code *

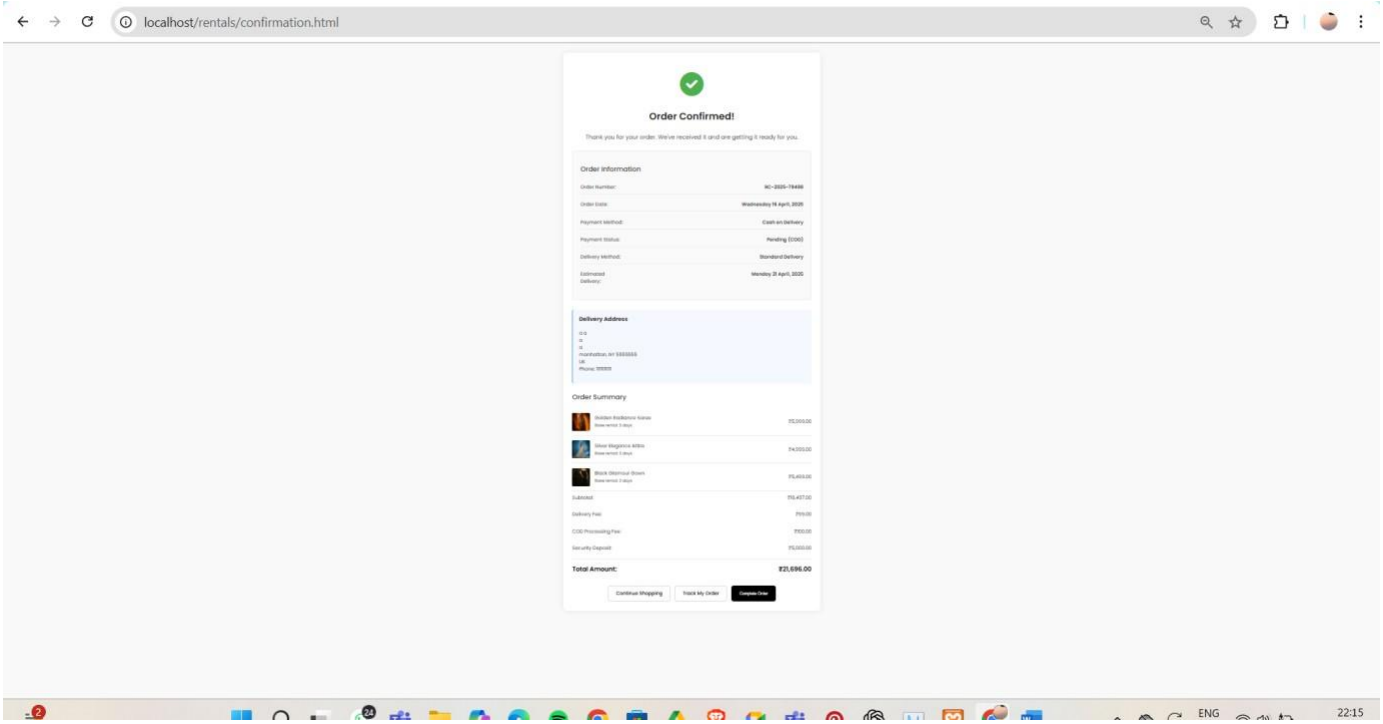
☐ Save this address for future orders

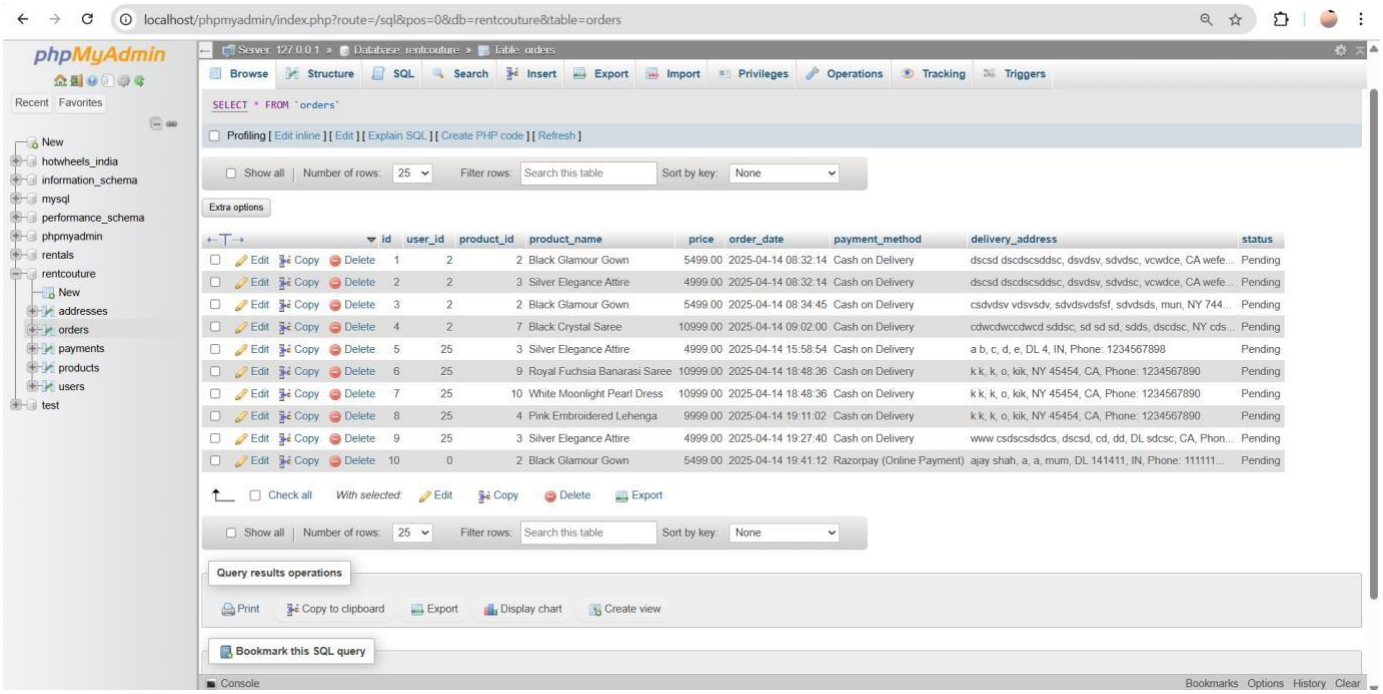
Back to cart

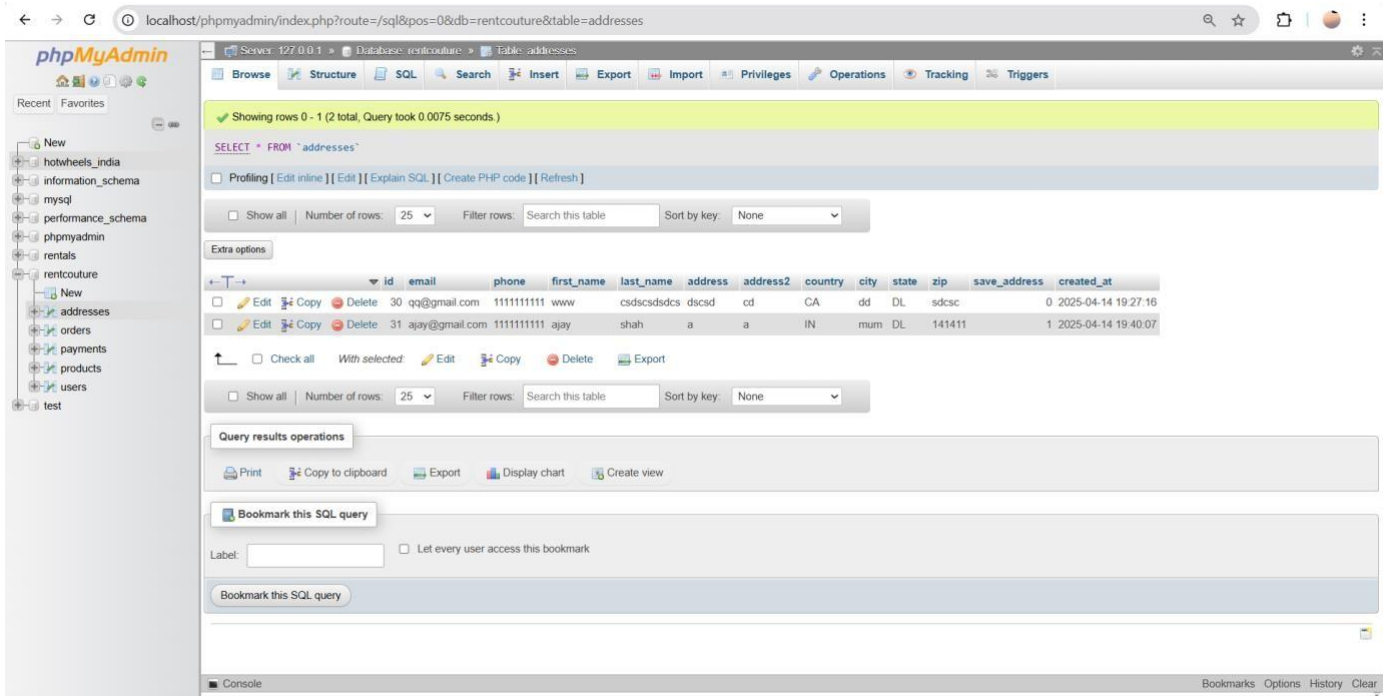
Continue to payment

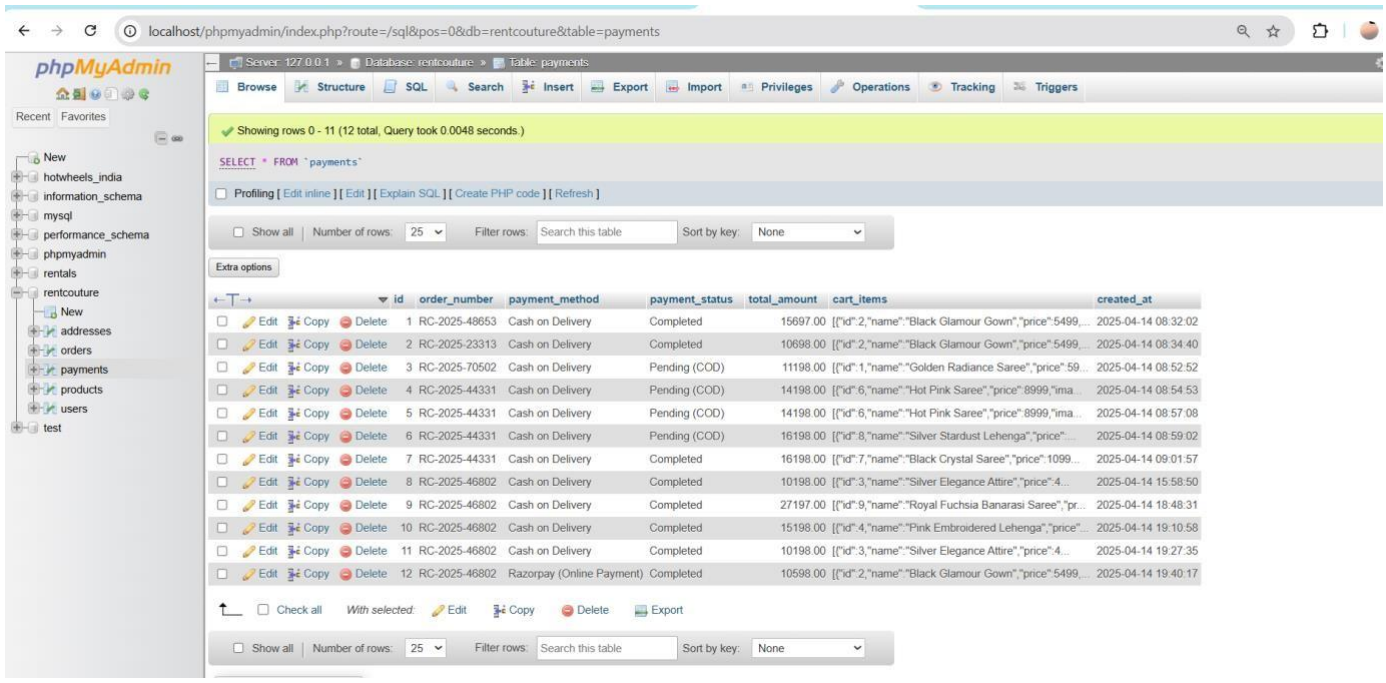

localhost/rentals/payment.html?order_id=4
Complete Your Purchase
Order Summary
Golden Radiance Saree (Base rental: 3 days) ₹5,999.00
Silver Elegance Attire (Base rental: 3 days) ₹4,999.00
Black Glamour Gown (Base rental: 3 days) ₹5,499.00
Subtotal: ₹16,497.00
Delivery Fee: ₹99.00
Security Deposit: ₹5,000.00
Total Amount: ₹21,596.00

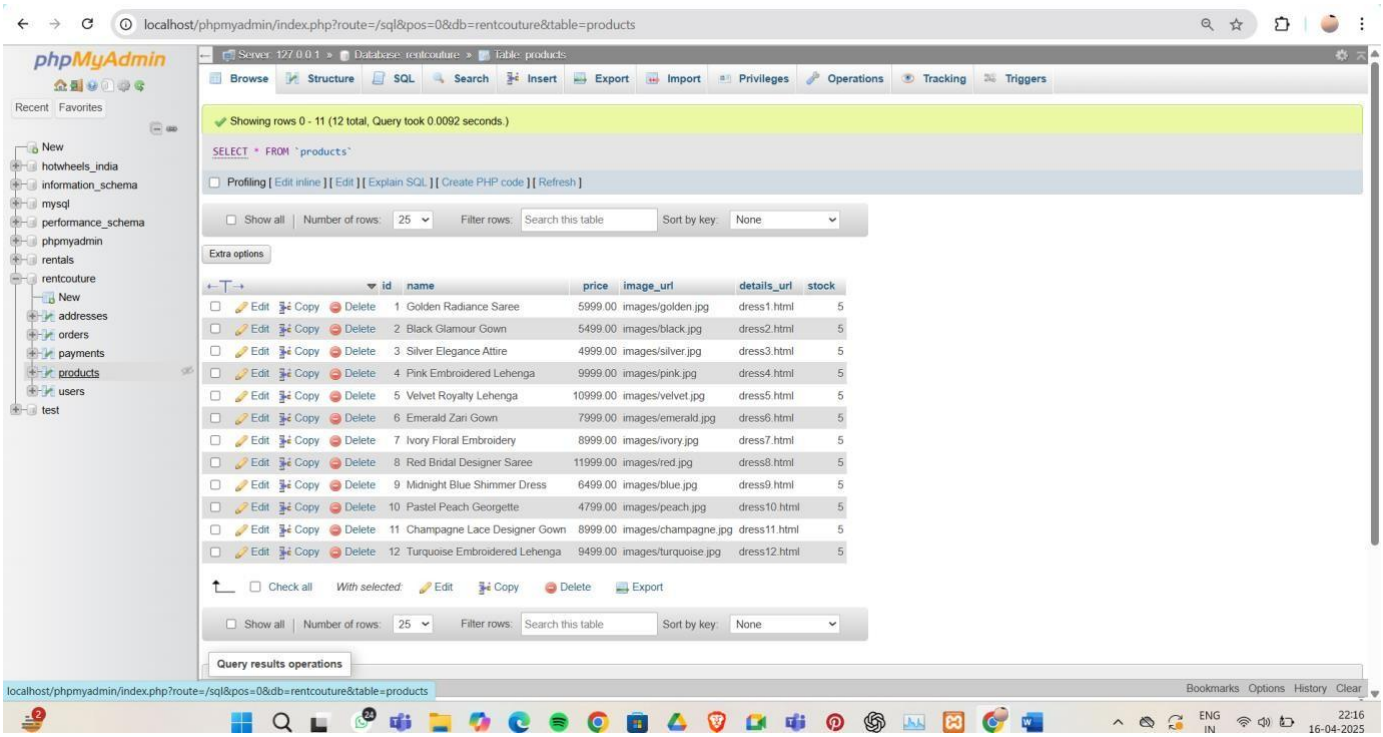

localhost/rentals/payment.html?order_id=4
Total Amount: ₹21,596.00
Select Payment Method
Credit/Debit Card, UPI, NetBanking (via Razorpay)
You'll be redirected to Razorpay's secure payment page after clicking "Pay Now"
Cash on Delivery
PAY NOW
Your payment information is processed securely. We do not store your credit card details.

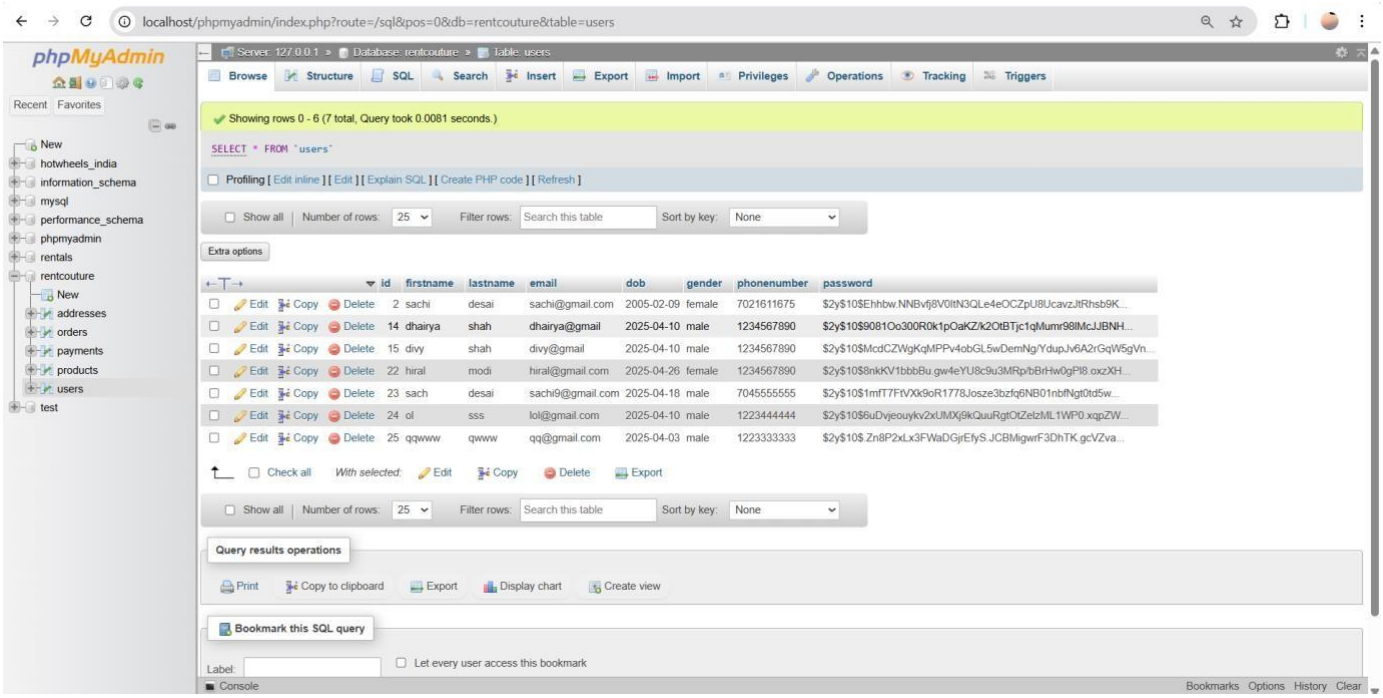

localhost/rentals/confirmation.html
Order Confirmed!
Order Information
Delivery Address
Order Summary
Total Amount:
₹21,696.00
Continue Shopping
Track My Order
Complete Order


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=rentcouture&table=orders
phpMyAdmin
Recent Favorites
New
hotwheels_india
information_schema
mysql
performance_schema
phpmyadmin
rentals
rentcouture
New
addresses
orders
payments
products
users
test
Server: 127.0.0.1 » Database: rentcouture » Table: orders
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
SELECT * FROM `orders`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id user_id product_id product_name price order_date payment_method delivery_address status
1 2 2 Black Glamour Gown 5499.00 2025-04-14 08:32:14 Cash on Delivery dscsd dscdscsddsc, dsvdsv, sdvdsc, vcwdce, CA wefe... Pending
2 2 3 Silver Elegance Attire 4999.00 2025-04-14 08:32:14 Cash on Delivery dscsd dscdscsddsc, dsvdsv, sdvdsc, vcwdce, CA wefe... Pending
3 2 2 Black Glamour Gown 5499.00 2025-04-14 08:34:45 Cash on Delivery csdvdsv vdsvsdv, sdvdsvdsfsf, sdvdsds, mun, NY 744... Pending
4 2 7 Black Crystal Saree 10999.00 2025-04-14 09:02:00 Cash on Delivery cdwcdwccdwcd sddsc, sd sd sd, sdds, dscdsc, NY cds... Pending
5 25 3 Silver Elegance Attire 4999.00 2025-04-14 15:58:54 Cash on Delivery a b, c, d, e, DL 4, IN, Phone: 1234567898 Pending
6 25 9 Royal Fuchsia Banarasi Saree 10999.00 2025-04-14 18:48:36 Cash on Delivery k k, k, o, kik, NY 45454, CA, Phone: 1234567890 Pending
7 25 10 White Moonlight Pearl Dress 10999.00 2025-04-14 18:48:36 Cash on Delivery k k, k, o, kik, NY 45454, CA, Phone: 1234567890 Pending
8 25 4 Pink Embroidered Lehenga 9999.00 2025-04-14 19:11:02 Cash on Delivery k k, k, o, kik, NY 45454, CA, Phone: 1234567890 Pending
9 25 3 Silver Elegance Attire 4999.00 2025-04-14 19:27:40 Cash on Delivery www csdscsdsdcs, dscsd, cd, dd, DL sdcsc, CA, Phon... Pending
10 0 2 Black Glamour Gown 5499.00 2025-04-14 19:41:12 Razorpay (Online Payment) ajay shah, a, a, mum, DL 141411, IN, Phone: 111111... Pending
Check all With selected: Edit Copy Delete Export
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Console
Bookmarks Options History Clear


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=rentcouture&table=addresses
phpMyAdmin
Recent Favorites
New
hotwheels_india
information_schema
mysql
performance_schema
phpmyadmin
rentals
rentcouture
New
addresses
orders
payments
products
users
test
Server: 127.0.0.1 » Database: rentcouture » Table: addresses
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
Showing rows 0 - 1 (2 total, Query took 0.0075 seconds.)
SELECT * FROM `addresses`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id email phone first_name last_name address address2 country city state zip save_address created_at
Edit Copy Delete 30 qq@gmail.com 1111111111 www csdscsdsdcs dscsd cd CA dd DL sdcsc 0 2025-04-14 19:27:16
Edit Copy Delete 31 ajay@gmail.com 1111111111 ajay shah a a IN mum DL 141411 1 2025-04-14 19:40:07
Check all With selected: Edit Copy Delete Export
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Bookmark this SQL query
Console
Bookmarks Options History Clear


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=rentcouture&table=payments
phpMyAdmin
Recent Favorites
New
hotwheels_india
information_schema
mysql
performance_schema
phpmyadmin
rentals
rentcouture
New
addresses
orders
payments
products
users
test
Server: 127.0.0.1 » Database: rentcouture » Table: payments
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
Showing rows 0 - 11 (12 total, Query took 0.0048 seconds.)
SELECT * FROM `payments`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id order_number payment_method payment_status total_amount cart_items created_at
Edit Copy Delete 1 RC-2025-48653 Cash on Delivery Completed 15697.00 [{"id":2,"name":"Black Glamour Gown","price":5499,... 2025-04-14 08:32:02
Edit Copy Delete 2 RC-2025-23313 Cash on Delivery Completed 10698.00 [{"id":2,"name":"Black Glamour Gown","price":5499,... 2025-04-14 08:34:40
Edit Copy Delete 3 RC-2025-70502 Cash on Delivery Pending (COD) 11198.00 [{"id":1,"name":"Golden Radiance Saree","price":59... 2025-04-14 08:52:52
Edit Copy Delete 4 RC-2025-44331 Cash on Delivery Pending (COD) 14198.00 [{"id":6,"name":"Hot Pink Saree","price":8999,"ima... 2025-04-14 08:54:53
Edit Copy Delete 5 RC-2025-44331 Cash on Delivery Pending (COD) 14198.00 [{"id":6,"name":"Hot Pink Saree","price":8999,"ima... 2025-04-14 08:57:08
Edit Copy Delete 6 RC-2025-44331 Cash on Delivery Pending (COD) 16198.00 [{"id":8,"name":"Silver Stardust Lehenga","price":... 2025-04-14 08:59:02
Edit Copy Delete 7 RC-2025-44331 Cash on Delivery Completed 16198.00 [{"id":7,"name":"Black Crystal Saree","price":1099... 2025-04-14 09:01:57
Edit Copy Delete 8 RC-2025-46802 Cash on Delivery Completed 10198.00 [{"id":3,"name":"Silver Elegance Attire","price":4... 2025-04-14 15:58:50
Edit Copy Delete 9 RC-2025-46802 Cash on Delivery Completed 27197.00 [{"id":9,"name":"Royal Fuchsia Banarasi Saree","pr... 2025-04-14 18:48:31
Edit Copy Delete 10 RC-2025-46802 Cash on Delivery Completed 15198.00 [{"id":4,"name":"Pink Embroidered Lehenga","price"... 2025-04-14 19:10:58
Edit Copy Delete 11 RC-2025-46802 Cash on Delivery Completed 10198.00 [{"id":3,"name":"Silver Elegance Attire","price":4... 2025-04-14 19:27:35
Edit Copy Delete 12 RC-2025-46802 Razorpay (Online Payment) Completed 10598.00 [{"id":2,"name":"Black Glamour Gown","price":5499,... 2025-04-14 19:40:17
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=rentcouture&table=products
phpMyAdmin
Recent Favorites
New
hotwheels_india
information_schema
mysql
performance_schema
phpmyadmin
rentals
rentcouture
New
addresses
orders
payments
products
users
test
Server: 127.0.0.1 » Database: rentcouture » Table: products
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
Showing rows 0 - 11 (12 total, Query took 0.0092 seconds.)
SELECT * FROM `products`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id name price image_url details_url stock
1 Golden Radiance Saree 5999.00 images/golden.jpg dress1.html 5
2 Black Glamour Gown 5499.00 images/black.jpg dress2.html 5
3 Silver Elegance Attire 4999.00 images/silver.jpg dress3.html 5
4 Pink Embroidered Lehenga 9999.00 images/pink.jpg dress4.html 5
5 Velvet Royalty Lehenga 10999.00 images/velvet.jpg dress5.html 5
6 Emerald Zari Gown 7999.00 images/emerald.jpg dress6.html 5
7 Ivory Floral Embroidery 8999.00 images/ivory.jpg dress7.html 5
8 Red Bridal Designer Saree 11999.00 images/red.jpg dress8.html 5
9 Midnight Blue Shimmer Dress 6499.00 images/blue.jpg dress9.html 5
10 Pastel Peach Georgette 4799.00 images/peach.jpg dress10.html 5
11 Champagne Lace Designer Gown 8999.00 images/champagne.jpg dress11.html 5
12 Turquoise Embroidered Lehenga 9499.00 images/turquoise.jpg dress12.html 5
Check all With selected: Edit Copy Delete Export
Query results operations
Bookmarks Options History Clear


localhost/phpmyadmin/index.php?route=/sql&pos=0&db=rentcouture&table=users
phpMyAdmin
Recent Favorites
New
hotwheels_india
information_schema
mysql
performance_schema
phpmyadmin
rentals
rentcouture
New
addresses
orders
payments
products
users
test
Server: 127.0.0.1 » Database: rentcouture » Table: users
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers
Showing rows 0 - 6 (7 total, Query took 0.0081 seconds.)
SELECT * FROM `users`
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
id firstname lastname email dob gender phonenumber password
2 sachi desai sachi@gmail.com 2005-02-09 female 7021611675 $2y$10$Ehhbw.NNBvfj8V0ltN3QLe4eOCZpU8UcavzJtRhsb9K...
14 dhairya shah dhairya@gmail 2025-04-10 male 1234567890 $2y$10$9081Oo300R0k1pOaKZ/k2OtBTjc1qMumr98lMcJJBNH...
15 divy shah divy@gmail 2025-04-10 male 1234567890 $2y$10$McdCZWgKqMPPv4obGL5wDemNg/YdupJv6A2rGqW5gVn...
22 hiral modi hiral@gmail.com 2025-04-26 female 1234567890 $2y$10$8nkKV1bbbBu.gw4eYU8c9u3MRp/bBrHw0gPl8.oxzXH...
23 sach desai sachi9@gmail.com 2025-04-18 male 7045555555 $2y$10$1mfT7FtVXk9oR1778Josze3bzfq6NB01nbfNgt0td5w...
24 ol sss lol@gmail.com 2025-04-10 male 1223444444 $2y$10$6uDvjeouykv2xUMXj9kQuuRgtOtZelzML1WP0.xqpZW...
25 qqwww qwww qq@gmail.com 2025-04-03 male 1223333333 $2y$10$.Zn8P2xLx3FWaDGjrEfyS.JCBMlgwrF3DhTK.gcVZva...
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print Copy to clipboard Export Display chart Create view
Bookmark this SQL query
Label:
Let every user access this bookmark
Console
Bookmarks Options History Clear

project:

1. **Database Design and Management**: We learned how to structure and organize complex databases to support real-world applications. Designing tables, creating relationships, and ensuring data integrity were essential components in building the backend of the platform.
2. **Full-Stack Development**: By developing both the front-end and back-end, we gained practical experience in handling both the client-side and server-side of web development. Using PHP for back-end and creating an intuitive UI for users allowed us to understand the complete flow of an ecommerce website.
3. **User Authentication and Security**: Implementing secure login systems, password encryption, and other security measures taught us the importance of safeguarding user data in an online platform. Understanding how to protect sensitive information is crucial in any real-world application.
4. **Handling Payments and Transactions**: Integrating payment gateways and ensuring smooth transaction flows was a key learning experience. We also gained insight into handling financial data and ensuring secure and reliable payment processing.
5. **UI/UX Design**: We focused on creating a user-friendly interface that caters to both fashion buyers and sellers. Understanding the importance of visual appeal and usability for an online platform helped us design an engaging experience for customers.
6. **Problem-Solving and Debugging**: While working on the project, we encountered numerous challenges related to features like product searches, rental durations, and payment integration. Debugging and resolving these issues improved our problem-solving and analytical thinking skills.
7. **Teamwork and Communication**: Collaboration was key to the success of the project. We learned how to divide tasks efficiently, communicate clearly within the team, and align everyone on the project vision. It also helped us understand the importance of teamwork in software development.
8. **Scalability and Performance Optimization**: As we worked on the platform, we had to ensure the system could handle a growing number of users and products. Learning how to optimize queries and improve the platform's performance was a valuable experience.

VIII. Learning from the Project

This project has been an incredible learning experience for me, providing valuable insights into both technical and practical aspects of developing a real-world application. Through the development of **RentCouture**, I gained hands-on experience in how databases work in a live environment, not just theoretically. My skills in **MySQL** and **PHP** improved significantly, especially in writing more efficient queries and managing data. The full-stack development process was a major learning curve, as I was involved in everything from designing the database schema to creating the user interface and integrating all components. I also focused on ensuring data validation and security, understanding the importance of safeguarding user information. Beyond the technical skills, the project taught me the importance of

structured planning, testing, and debugging, making the development process more efficient. Ultimately, it has provided me with a comprehensive understanding of how various software components come together to create a fully functional web application.

IX. Challenges Faced

During the development of **RentCouture**, we encountered several technical and logical challenges. Some of the major ones were:

- **Real-Time Product Availability Updates**: One of the biggest challenges was ensuring that product availability was updated in real-time. When a user rented a product, the availability had to be immediately reflected for all users, which required implementing efficient backend logic and handling asynchronous updates.
- **Handling Multiple Rentals Simultaneously**: Managing multiple users renting the same product within a short time span was tricky. We had to write logic to ensure that product availability was checked properly, preventing double bookings and maintaining accurate stock levels.
- **Payment Gateway Integration**: Integrating a secure payment gateway was challenging, as we had to ensure that transactions were processed smoothly, and users were informed of successful or failed payments in real time. This required learning the intricacies of secure API integration and handling transaction errors effectively.
- **Database Connectivity**: We faced issues connecting the frontend with the MySQL database, especially when dealing with complex queries for filtering products and handling user data. Overcoming this required a deeper understanding of how HTTP requests and database queries worked together.
- **UI/UX Design**: Designing an intuitive and aesthetically pleasing interface was another challenge. We wanted to ensure that users could easily navigate the platform, search for products, place orders, and manage rentals seamlessly, all while keeping the design clean and responsive across devices.

Despite these challenges, we worked collaboratively, researched solutions, and successfully resolved the issues through constant learning and iteration.

XI. Conclusion

- In conclusion, the development of **RentCouture** provided us with valuable practical experience in creating a full-stack web application. We successfully built a fashion rental platform that allows users to register, browse products, rent fashion items, and make secure payments.
- The project gave us a deeper understanding of how the different components of a web application— frontend design, backend logic, database management, and payment gateway integration— work together seamlessly. It also enhanced our skills in problem-solving, teamwork, and project management.

- This hands-on experience boosted my confidence in tackling more complex projects in the future and reinforced my knowledge of web development. Overall, **RentCouture** was both a challenging and rewarding experience, significantly contributing to our technical expertise and personal growth.